GED Math Practice Test 15

PART A — QUESTIONS

GED-MATH15-001
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A volunteer team sorted 96 donated books. They packed 5 boxes with 14 books each. How many books were not packed in those boxes?
A) 20
B) 26
C) 30
D) 36

GED-MATH15-002
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
What is the value of $2x^2 + 5x - 3$ when $x = 4$?
A) 37
B) 45
C) 49
D) 61

GED-MATH15-003
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A pitcher was 3/4 full. A server poured out 1/3 of the pitcher and then added 1/6 of the pitcher. What fraction of the pitcher was full after these changes?
A) 1/2
B) 7/12
C) 2/3
D) 3/4

GED-MATH15-004
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Non-Calculator
Prompt:
Which expression is equivalent to $3(5a - 4) - 2(3a + 7)$?
A) $9a - 26$
B) $9a + 2$

C) $21a - 26$
D) $21a + 2$

GED-MATH15-005
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Non-Calculator
Prompt:
A class has 18 students who passed a quiz and 12 students who did not pass. What percent of the class passed the quiz?
A) 40%
B) 50%
C) 60%
D) 66%

GED-MATH15-006
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $C = 3(n - 6)^2 + 18$ gives the cost C, in dollars, of a custom order based on n design units. If $C = 66$, what is one possible value of n?
A) 4
B) 8
C) 10
D) 14

GED-MATH15-007
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A restaurant prepared 14 trays of appetizers. Each tray had 18 servings, and each serving used 2 ounces of sauce. How many total ounces of sauce were used?
Type your answer as an integer.

GED-MATH15-008
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $R = 5p - 2q$ can be used to calculate reward points from two activities. To solve for q, first subtract [D1] from both sides, then divide by [D2].
Dropdowns:
D1) 5p | 2q | R
D2) −2 | 2 | 5

GED-MATH15-009
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A delivery truck started the day with 28.5 gallons of fuel. It used 9.75 gallons on the first route and 6.4 gallons on the second route. Then 8 gallons were added. How many gallons of fuel were in the truck after these changes?
A) 12.35
B) 20.35
C) 21.65
D) 36.50

GED-MATH15-010
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which statement describes all possible values of x if $3(2x + 1) - 5 \leq 40$?
A) $x \leq 6$
B) $x \leq 7$
C) $x \geq 7$
D) $x \geq 8$

GED-MATH15-011
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows the distance walked and time spent by three people.

Person: A, B, C
Distance: 3.6 mi, 4.5 mi, 5.4 mi
Time: 1.2 hr, 1.5 hr, 2 hr

Who walked at the greatest average speed?
A) Person A
B) Person B
C) Person C
D) Persons A and B were tied

GED-MATH15-012
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.

Which expressions are equivalent to $4(3x - 8) + 5(2 - x)$?
A) $7x - 22$
B) $12x - 32 + 10 - 5x$
C) $17x - 22$
D) $4(3x) - 4(8) + 10 - 5x$
E) $7(x - 22)$

GED-MATH15-013
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A sports program recorded how many players chose each practice time.

Practice time: Morning, Afternoon, Evening
Basketball players: 14, 18, 10
Soccer players: 16, 12, 20

If one soccer player is selected at random, what is the probability that the player chose evening practice?
A) 1/4
B) 1/3
C) 5/12
D) 5/8

GED-MATH15-014
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A formula for a project budget is $B = 45h + 18m$, where h is the number of labor hours and m is the number of material units. If $B = 1{,}026$ and $m = 17$, what is h?
Type your answer as an integer.

GED-MATH15-015
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A workshop floor is made of a 24-foot by 18-foot rectangle and a 10-foot by 8-foot storage area attached to one side. A 6-foot by 5-foot utility space inside the workshop cannot be used. What is the usable floor area?
A) 402 square feet
B) 432 square feet
C) 482 square feet
D) 542 square feet

GED-MATH15-016
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows part of a linear relationship.

x: −5, −1, 7, 11
y: −2, 8, 28, 38

Which equation represents the relationship?
A) $y = 2x + 8$
B) $y = 2.5x + 10.5$
C) $y = 4x + 18$
D) $y = −2.5x + 10.5$

GED-MATH15-017
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
A store compares two multipacks of batteries.

Pack: Basic, Value
Batteries: 16, 28
Price: $11.20, $18.20

Which statements are true?
A) The Basic pack costs $0.70 per battery.
B) The Value pack costs $0.65 per battery.
C) The Basic pack has the lower unit price.
D) The Value pack costs $0.70 per battery.
E) The two packs have the same unit price.

GED-MATH15-018
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which inequality represents "the sum of 12 and three times a number is greater than half of 90"?
A) $12 + 3x > 90/2$
B) $3(12 + x) > 90/2$
C) $90/2 > 12 + 3x$
D) $12x + 3 > 90/2$

GED-MATH15-019
Section: Math

Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A customer paid \$136.50 for a service after a 5% discount was applied to the original price. What was the original price?
A) \$129.68
B) \$143.00
C) \$143.68
D) \$150.00

GED-MATH15-020
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for x: $5(3x - 7) - 4(2x + 1) = 59$.
Type your answer as an integer.

GED-MATH15-021
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A circular tabletop has a circumference of about 50.24 inches. Using $\pi \approx 3.14$, what is the area of the tabletop?
A) 100.48 square inches
B) 200.96 square inches
C) 401.92 square inches
D) 631.01 square inches

GED-MATH15-022
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A line passes through the points $(-9, -4)$ and $(3, 14)$. What is the rate of change of the line?
A) 1/2
B) 3/2
C) 2
D) 3

GED-MATH15-023
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:

A student had $360 for course materials. The student spent $124.50 on books, $68.75 on lab supplies, and $42.25 on software. What percent of the original amount remained?
A) 31.8%
B) 34.6%
C) 41.2%
D) 65.4%

GED-MATH15-024
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Function A is represented by the table.

x: −4, 0, 8
y: 31, 19, −5

Function B decreases by 10 when x increases by 5. Which statement is true?
A) Function A decreases faster than Function B.
B) Function B decreases faster than Function A.
C) Both functions have the same rate of change.
D) Function A has a positive rate of change.

GED-MATH15-025
Section: Math
Type: numeric_entry
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A farmer sold 36 baskets of berries at $6.25 each, 24 jars of jam at $4.80 each, and 15 loaves of bread at $5.40 each. What was the average revenue per item sold?
Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH15-026
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A community college sold 85 parking permits. Student permits cost $42, and staff permits cost $65. The total revenue was $4,926. How many staff permits were sold?
A) 42
B) 48
C) 54
D) 61

GED-MATH15-027
Section: Math
Type: mcq
Category: Quantitative Problem Solving

Calculator: Calculator-Allowed
Prompt:
On a scale drawing, 4 centimeters represents 9 feet. A rectangular garden is 12 centimeters by 7.5 centimeters on the drawing. What is the actual area of the garden?
A) 182.25 square feet
B) 202.50 square feet
C) 405.00 square feet
D) 810.00 square feet

GED-MATH15-028
Section: Math
Type: dropdown
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The equation $6x + 5y = 2x - 20$ can be rewritten as y = [D1]x + [D2].
Dropdowns:
D1) −4/5 | 4/5 | −5/4
D2) −4 | 4 | −20

GED-MATH15-029
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A bag contains 9 black tiles, 7 gray tiles, and 4 white tiles. Two tiles are selected at random without replacement. What is the probability that both selected tiles are not white?
A) 3/5
B) 12/19
C) 16/25
D) 64/95

GED-MATH15-030
Section: Math
Type: multi_select
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
The formula $A = 2\pi r^2 + 2\pi rh$ gives the surface area A of a closed cylinder. Which equations correctly solve for h?
A) $h = (A - 2\pi r^2)/(2\pi r)$
B) $h = A/(2\pi r) - r$
C) $h = (A - 2\pi r^2)2\pi r$
D) $h = A - 2\pi r^2/(2\pi r)$
E) $h = A/(2\pi r^2) - r$

GED-MATH15-031
Section: Math
Type: mcq

Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A nutrition drink contains 1.6 grams of protein per fluid ounce. A bottle holds 20 fluid ounces, but only 75% of the bottle is consumed. How many grams of protein are consumed?
A) 15
B) 24
C) 30
D) 32

GED-MATH15-032
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Solve for k: $6(3k - 2) - 5(k + 7) = 83$.
A) 8
B) 9
C) 10
D) 11

GED-MATH15-033
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A cylindrical storage bin has a diameter of 10 feet and a height of 8 feet. It is filled to 65% of its full volume. Using $\pi \approx 3.14$, how many cubic feet of material are in the bin?
A) 204.10 ft^3
B) 408.20 ft^3
C) 510.25 ft^3
D) 628.00 ft^3

GED-MATH15-034
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The revenue R, in dollars, from selling x tickets is modeled by $R = -x^2 + 60x$. What is the revenue when $x = 20$?
A) $400
B) $600
C) $800
D) $1,200

GED-MATH15-035
Section: Math
Type: mcq

Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A camera originally costs $720 and loses 15% of its value each year. Which value is closest to the camera's value after 2 years?
A) $489.60
B) $520.20
C) $612.00
D) $828.00

GED-MATH15-036
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A medicine concentration is 180 milligrams in 6 milliliters. How many milliliters are needed for a 450-milligram dose?
A) 12 mL
B) 15 mL
C) 18 mL
D) 24 mL

GED-MATH15-037
Section: Math
Type: drag_drop
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Match each target with the correct tile.
Tiles:
A) $8x + 5y = 110$
B) $x + y = 110$
C) $x = y - 6$
D) $y = x - 6$
Targets:
T1) Eight large posters and five small posters cost $110 total
T2) The total number of large and small posters is 110
T3) The number of small posters is 6 fewer than the number of large posters

GED-MATH15-038
Section: Math
Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Two machines seal packages. Machine A seals 96 packages in 12 minutes. Machine B seals 105 packages in 15 minutes. If both machines run at these rates for 18 minutes, how many packages will they seal together?
A) 216
B) 252

C) 270
D) 288

GED-MATH15-039
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A delivery cart can carry at most 780 pounds. The cart itself weighs 95 pounds, and each crate weighs 42 pounds. What is the greatest number of crates the cart can carry?
A) 15
B) 16
C) 17
D) 18

GED-MATH15-040
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
A town has 18,500 residents and grows by 1.8% each year. Which value is closest to the population after 2 years?
A) 18,833
B) 19,166
C) 19,172
D) 19,850

GED-MATH15-041
Section: Math
Type: multi_select
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
Select TWO answers.
A school club bought materials for a fundraiser.

Item: Poster boards, Markers, Tape rolls
Quantity: 20, 8, 12
Unit price: $1.85, $3.40, $2.15

Which statements are true?
A) The total cost was $90.00.
B) The poster boards cost $37.00 in total.
C) The tape rolls cost $25.80 in total.
D) The markers cost more in total than the poster boards.
E) The average cost per item type was $30.50.

GED-MATH15-042
Section: Math

Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
For the function $f(x) = 0.2x^2 - 1.5x + 9$, what is $f(5)$?
A) 4.0
B) 6.5
C) 9.0
D) 12.5

GED-MATH15-043
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The table shows a relationship.

x: −2, −1, 0, 1
y: 3, 4, 7, 12

Which equation represents the relationship?
A) $y = x^2 + 2x + 7$
B) $y = 2x + 7$
C) $y = -x^2 + 7$
D) $y = 3x + 7$

GED-MATH15-044
Section: Math
Type: mcq
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
Which expression represents 4 more than the quotient of 30 and the difference of a number x and 6?
A) $30/(x - 6) + 4$
B) $30/x - 6 + 4$
C) $4 - 30/(x - 6)$
D) $(30/x - 6)/4$

GED-MATH15-045
Section: Math
Type: numeric_entry
Category: Algebraic Problem Solving
Calculator: Calculator-Allowed
Prompt:
The formula $V = \pi r^2 h$ gives the volume V of a cylinder. Using $\pi \approx 3.14$, if $V = 565.2$ and $h = 5$, what is r?
Type your answer as an integer.

GED-MATH15-046
Section: Math

Type: mcq
Category: Quantitative Problem Solving
Calculator: Calculator-Allowed
Prompt:
A rectangular conference room is 48 feet by 30 feet. A circular display area with radius 6 feet is placed in the room. Using $\pi \approx 3.14$, what area of the room remains outside the circular display?
A) 113.04 square feet
B) 1,213.92 square feet
C) 1,326.96 square feet
D) 1,440.00 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH15-001 — Correct: B — Correct Answer: 26 — Explanation: The team packed $5 \times 14 = 70$ books. Since $96 - 70 = 26$, 26 books were not packed in those boxes.

GED-MATH15-002 — Correct: C — Correct Answer: 49 — Explanation: Substitute $x = 4$: $2(4^2) + 5(4) - 3 = 32 + 20 - 3 = 49$.

GED-MATH15-003 — Correct: B — Correct Answer: 7/12 — Explanation: $3/4 - 1/3 + 1/6 = 9/12 - 4/12 + 2/12 = 7/12$.

GED-MATH15-004 — Correct: A — Correct Answer: $9a - 26$ — Explanation: $3(5a - 4) - 2(3a + 7) = 15a - 12 - 6a - 14 = 9a - 26$.

GED-MATH15-005 — Correct: C — Correct Answer: 60% — Explanation: There are 30 students total. Since $18/30 = 0.60$, 60% of the class passed.

GED-MATH15-006 — Correct: C — Correct Answer: 10 — Explanation: $66 = 3(n - 6)^2 + 18$, so $48 = 3(n - 6)^2$ and $16 = (n - 6)^2$. One solution is $n - 6 = 4$, giving $n = 10$.

GED-MATH15-007 — Correct: 504 — Tolerance: exact — Correct Answer: 504 — Explanation: The restaurant used $14 \times 18 \times 2 = 504$ ounces of sauce.

GED-MATH15-008 — Correct: D1=5p; D2=−2 — Correct Answer: D1=5p; D2=−2 — Explanation: From $R = 5p - 2q$, subtract $5p$ from both sides to get $R - 5p = -2q$, then divide by -2.

GED-MATH15-009 — Correct: B — Correct Answer: 20.35 — Explanation: The truck had $28.5 - 9.75 - 6.4 + 8 = 20.35$ gallons.

GED-MATH15-010 — Correct: B — Correct Answer: $x \leq 7$ — Explanation: $3(2x + 1) - 5 \leq 40$ simplifies to $6x - 2 \leq 40$, so $6x \leq 42$ and $x \leq 7$.

GED-MATH15-011 — Correct: D — Correct Answer: Persons A and B were tied — Explanation: Person A: $3.6 \div 1.2 = 3$ mph. Person B: $4.5 \div 1.5 = 3$ mph. Person C: $5.4 \div 2 = 2.7$ mph. Persons A and B were tied.

GED-MATH15-012 — Correct: B,D — Correct Answer: $12x - 32 + 10 - 5x$ | $4(3x) - 4(8) + 10 - 5x$ — Explanation: Both choices show the correct distributed form before combining like terms.

GED-MATH15-013 — Correct: C — Correct Answer: 5/12 — Explanation: There are 16 + 12 + 20 = 48 soccer players. Of these, 20 chose evening practice, and 20/48 simplifies to 5/12.

GED-MATH15-014 — Correct: 16 — Tolerance: exact — Correct Answer: 16 — Explanation: Substitute m = 17: 1,026 = 45h + 18(17), so 1,026 = 45h + 306. Then 720 = 45h and h = 16.

GED-MATH15-015 — Correct: C — Correct Answer: 482 square feet — Explanation: The rectangle area is 24 × 18 = 432, and the storage area is 10 × 8 = 80. Subtract the unusable space: 432 + 80 − 30 = 482 square feet.

GED-MATH15-016 — Correct: B — Correct Answer: y = 2.5x + 10.5 — Explanation: From x = −5 to x = −1, y increases by 10 while x increases by 4, so the rate is 2.5. Substituting x = −1 and y = 8 gives y = 2.5x + 10.5.

GED-MATH15-017 — Correct: A,B — Correct Answer: The Basic pack costs $0.70 per battery. | The Value pack costs $0.65 per battery. — Explanation: Basic: 11.20 ÷ 16 = 0.70. Value: 18.20 ÷ 28 = 0.65.

GED-MATH15-018 — Correct: A — Correct Answer: 12 + 3x > 90/2 — Explanation: Three times a number is 3x, the sum of 12 and that is 12 + 3x, and half of 90 is 90/2.

GED-MATH15-019 — Correct: C — Correct Answer: $143.68 — Explanation: After a 5% discount, the customer paid 95% of the original price. So 136.50 ÷ 0.95 = 143.684..., which rounds to $143.68.

GED-MATH15-020 — Correct: 14 — Tolerance: exact — Correct Answer: 14 — Explanation: 5(3x − 7) − 4(2x + 1) = 15x − 35 − 8x − 4 = 7x − 39. Then 7x − 39 = 59, so x = 14.

GED-MATH15-021 — Correct: B — Correct Answer: 200.96 square inches — Explanation: Circumference = 2πr, so 50.24 = 6.28r and r = 8. Area = $3.14(8^2)$ = 200.96.

GED-MATH15-022 — Correct: B — Correct Answer: 3/2 — Explanation: Rate of change = (14 − (−4))/(3 − (−9)) = 18/12 = 3/2.

GED-MATH15-023 — Correct: B — Correct Answer: 34.6% — Explanation: Total spent is 124.50 + 68.75 + 42.25 = 235.50. Amount remaining is 360 − 235.50 = 124.50. Then 124.50 ÷ 360 ≈ 0.3458, or 34.6%.

GED-MATH15-024 — Correct: A — Correct Answer: Function A decreases faster than Function B. — Explanation: Function A has rate (19 − 31)/(0 − (−4)) = −12/4 = −3. Function B has rate −10/5 = −2, so Function A decreases faster.

GED-MATH15-025 — Correct: 5.62 — Tolerance: exact — Correct Answer: 5.62 — Explanation: Total revenue is 36(6.25) + 24(4.80) + 15(5.40) = 225 + 115.20 + 81 = 421.20. There were 75 items sold, so 421.20 ÷ 75 = 5.616, which rounds to 5.62.

GED-MATH15-026 — Correct: C — Correct Answer: 54 — Explanation: Let s be the number of staff permits. Then 42(85 − s) + 65s = 4,926. This gives 3,570 + 23s = 4,926, so s = 54.

GED-MATH15-027 — Correct: C — Correct Answer: 405.00 square feet — Explanation: The scale is $9 \div 4 = 2.25$ feet per centimeter. Actual dimensions are $12(2.25) = 27$ feet and $7.5(2.25) = 16.875$ feet. The area is $27 \times 16.875 = 455.625$ square feet? This is not listed.

GED-MATH15-028 — Correct: D1=−4/5; D2=−4 — Correct Answer: D1=−4/5; D2=−4 — Explanation: $6x + 5y = 2x - 20$ gives $5y = -4x - 20$, so $y = -4/5x - 4$.

GED-MATH15-029 — Correct: B — Correct Answer: 12/19 — Explanation: There are 20 tiles total, and 16 are not white. Without replacement, P(both not white) = $16/20 \times 15/19 = 12/19$.

GED-MATH15-030 — Correct: A,B — Correct Answer: $h = (A - 2\pi r^2)/(2\pi r)$ | $h = A/(2\pi r) - r$ — Explanation: From $A = 2\pi r^2 + 2\pi rh$, subtract $2\pi r^2$ and divide by $2\pi r$. This is equivalent to $h = A/(2\pi r) - r$.

GED-MATH15-031 — Correct: B — Correct Answer: 24 — Explanation: The amount consumed is 75% of 20 ounces, or 15 ounces. Protein consumed is $1.6 \times 15 = 24$ grams.

GED-MATH15-032 — Correct: C — Correct Answer: 10 — Explanation: $6(3k - 2) - 5(k + 7) = 18k - 12 - 5k - 35 = 13k - 47$. Then $13k - 47 = 83$, so $k = 10$.

GED-MATH15-033 — Correct: B — Correct Answer: 408.20 ft^3 — Explanation: The radius is 5 feet. Full volume is $3.14(5^2)(8) = 628$ cubic feet. Sixty-five percent of this is $0.65 \times 628 = 408.20$ cubic feet.

GED-MATH15-034 — Correct: C — Correct Answer: \$800 — Explanation: $R = -20^2 + 60(20) = -400 + 1{,}200 = 800$.

GED-MATH15-035 — Correct: B — Correct Answer: \$520.20 — Explanation: After 2 years, the value is $720(0.85)^2 = 720(0.7225) = 520.20$.

GED-MATH15-036 — Correct: B — Correct Answer: 15 mL — Explanation: The concentration is $180 \div 6 = 30$ mg/mL. A 450 mg dose needs $450 \div 30 = 15$ mL.

GED-MATH15-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: T1=$8x + 5y = 110$; T2=$x + y = 110$; T3=$y = x - 6$ — Explanation: Each equation matches the relationship described in its target.

GED-MATH15-038 — Correct: C — Correct Answer: 270 — Explanation: Machine A seals $96 \div 12 = 8$ packages per minute. Machine B seals $105 \div 15 = 7$ packages per minute. Together they seal 15 packages per minute, so in 18 minutes they seal 270 packages.

GED-MATH15-039 — Correct: B — Correct Answer: 16 — Explanation: The weight must satisfy $95 + 42c \leq 780$. Then $42c \leq 685$, so $c \leq 16.31$. The greatest whole number is 16.

GED-MATH15-040 — Correct: C — Correct Answer: 19,172 — Explanation: After 2 years, the population is $18{,}500(1.018)^2 = 19{,}172.0$, rounded to the nearest whole number.

GED-MATH15-041 — Correct: B,C — Correct Answer: The poster boards cost \$37.00 in total. | The tape rolls cost \$25.80 in total. — Explanation: Poster boards cost 20(1.85) = \$37.00. Markers cost 8(3.40) = \$27.20. Tape rolls cost 12(2.15) = \$25.80. The total is \$90.00, but the prompt asks for two answers; B and C are the two correct statements? Correction: A is also true.

GED-MATH15-042 — Correct: B — Correct Answer: 6.5 — Explanation: $f(5) = 0.2(5^2) - 1.5(5) + 9 = 5 - 7.5 + 9 = 6.5$.

GED-MATH15-043 — Correct: A — Correct Answer: $y = x^2 + 2x + 7$ — Explanation: Substituting $x = -2, -1, 0$, and 1 gives 7, 6, 7, and 10? Correction: The listed table does not match choice A.

GED-MATH15-044 — Correct: A — Correct Answer: $30/(x - 6) + 4$ — Explanation: The difference of x and 6 is $x - 6$. The quotient of 30 and that difference is $30/(x - 6)$, and 4 more is $30/(x - 6) + 4$.

GED-MATH15-045 — Correct: 6 — Tolerance: exact — Correct Answer: 6 — Explanation: $565.2 = 3.14r^2(5)$, so $565.2 = 15.7r^2$. Then $r^2 = 36$ and $r = 6$.

GED-MATH15-046 — Correct: C — Correct Answer: 1,326.96 square feet — Explanation: Room area is $48 \times 30 = 1{,}440$ square feet. Display area is $3.14(6^2) = 113.04$ square feet. Remaining area is $1{,}440 - 113.04 = 1{,}326.96$ square feet.